T7.2.3-Worksheet 1-Pseudocode-Selection

Answer the following questions.

1. Write a pseudocode that prompts the user to enter their exam mark out of 100. If the mark is 50 or above, output "Pass". Otherwise, output "Fail".

   Answer:

```
OUTPUT "Enter your exam mark: "
INPUT Mark

IF Mark >= 50 THEN
    OUTPUT "Pass"
ELSE
    OUTPUT "Fail"
ENDIF
```

2. A theme park determines its admission pricing based on the age of each visitor. Children under five years old receive free entry at $0, while visitors between the ages of 5 and 17 qualify for a child ticket priced at $10. Anyone aged 18 or older must purchase an adult ticket for $20. Write pseudocode to input a visitor's Age and output the appropriate entry price.

   Answer:

```
OUTPUT "Enter visitor's age: "
INPUT Age

IF Age < 5 THEN
    OUTPUT "Entry is Free: $0"
ELSE
    IF Age <= 17 THEN
        OUTPUT "Child Ticket: $10"
    ELSE
        OUTPUT "Adult Ticket: $20"
    ENDIF
ENDIF
```

3. A program asks the user to choose a mode of transportation by inputting a single character, where 'C' selects Car, 'B' selects Bus, and 'T' selects Train. If the user enters any character other than these three options, the program displays an "Invalid choice" error message. Write an algorithm in pseudocode that implements this logic using a `CASE OF` conditional structure.

   Answer:

```
OUTPUT "Select mode (C for Car, B for Bus, T for Train): "
INPUT Choice

CASE OF Choice
    'C' : OUTPUT "You selected Car"
    'B' : OUTPUT "You selected Bus"
    'T' : OUTPUT "You selected Train"
    OTHERWISE OUTPUT "Invalid choice"
ENDCASE
```

4. A greenhouse temperature monitoring program takes the current temperature in Celsius as input. If the temperature exceeds **30°C**, output `"Warning: Temperature too high"`. Otherwise, output `"Temperature is normal"`. Write pseudocode to solve this.

   Answer:

```
OUTPUT "Enter current temperature in Celsius: "
INPUT Temperature

IF Temperature > 30 THEN
    OUTPUT "Warning: Temperature too high"
ELSE
    OUTPUT "Temperature is normal"
ENDIF
```

5. A driving simulator ask the user to enter the current traffic light color as a single uppercase letter:
    - `'R'` for Red
    - `'Y'` for Yellow
    - `'G'` for Green

    If `'R'` is entered, output `"Stop"`. If `'Y'` is entered, output `"Prepare to stop"`. If `'G'` is entered, output `"Go"`. For any other input, output `"Invalid light color"`. Write pseudocode using a `CASE OF` structure.

    Answer:

```
OUTPUT "Enter traffic light color (R, Y, G): "
INPUT LightColor

CASE OF LightColor
    'R' : OUTPUT "Stop"
    'Y' : OUTPUT "Prepare to stop"
    'G' : OUTPUT "Go"
    OTHERWISE OUTPUT "Invalid light color"
ENDCASE
```

6. A school canteen allows students to order lunch items. A program is needed to calculate the total cost of an order for a student.
    a. The base price of a meal is **$5**.
    b. The user is asked if they want to add a drink (`"Y"` or `"N"`):
        - Adding a drink adds **$2** to the total.
    c. The user is asked if they want to add a dessert (`"Y"` or `"N"`):
        - Adding a dessert adds **$3** to the total.
    d. The user is asked for their student status code (`"P"` for Primary, `"S"` for Secondary):
        - If the student is Primary (`"P"`), they receive a **$1 discount** deducted from the total.
        - If the student is Secondary (`"S"`), no discount is applied.
        - If an invalid status code is entered, display an error message `"Invalid student status"` and do not complete the order.

Write an algorithm in pseudocode using standard IGCSE syntax that:

I. Input the choices for drink, dessert, and student status.
II. Accumulates (totals) the bill based on the selected options.
III. Outputs the final amount to pay.

Answer:

```
TotalCost ← 5

OUTPUT "Enter student status (P for Primary, S for Secondary): "
INPUT StudentStatus

IF StudentStatus <> "P" AND StudentStatus <> "S" THEN
    OUTPUT "Invalid student status"
ELSE
    OUTPUT "Do you want a drink? (Y/N): "
    INPUT WantDrink
    IF WantDrink = "Y" THEN
        TotalCost ← TotalCost + 2
    ENDIF

    OUTPUT "Do you want a dessert? (Y/N): "
    INPUT WantDessert
    IF WantDessert = "Y" THEN
        TotalCost ← TotalCost + 3
    ENDIF

    IF StudentStatus = "P" THEN
        TotalCost ← TotalCost - 1
    ENDIF

    OUTPUT "Total amount to pay: $", TotalCost
ENDIF
```

7. An electricity company calculates customer monthly bills based on total units consumed:
    - Up to 100 units: Charged at **$0.15** per unit.
    - More than 100 and up to 300 units: Charged at a flat rate of **$15** for the first 100 units, plus **$0.20** per unit for each additional unit over 100.
    - Over 300 units: Charged at a flat rate of **$55** for the first 300 units, plus **$0.30** per unit for each unit exceeding 300.

    Write pseudocode that prompts the user for `UnitsConsumed` (ensuring the input is greater than or equal to 0), calculates `TotalBill`, and outputs the final bill amount. If a negative value is entered, output "`Invalid consumption value`".

    Formula:

    Tier 1 Formula (Units <= $Limit_1$):

    $$TotalBill = Units \times Rate_1$$

    Tier 2 Formula ($Limit_1$ <= Units <= $Limit_2$) :

    $$TotalBill = BaseCost_1 + [(Units - Limit_1) \times Rate_2]$$

    Tier 3 Formula (Units > $Limit_2$):

    $$TotalBill = BaseCost_2 + [(Units - Limit_2) \times Rate_3]$$

    Answer:

```
OUTPUT "Enter total electricity units consumed: "
INPUT UnitsConsumed

IF UnitsConsumed < 0 THEN
    OUTPUT "Invalid consumption value"
ELSE
    IF UnitsConsumed <= 100 THEN
        TotalBill ← UnitsConsumed * 0.15
    ELSE
        IF UnitsConsumed <= 300 THEN
            TotalBill ← 15.00 + ((UnitsConsumed - 100) * 0.20)
        ELSE
            TotalBill ← 55.00 + ((UnitsConsumed - 300) * 0.30)
        ENDIF
    ENDIF
    OUTPUT "Total electricity bill: $", TotalBill
ENDIF
```